AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 5 · STUDENT LESSON

Inspect Before Modeling

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Keep the raw table. Count the flags. Explain the overlap.

Inspect data and document provenance without inventing repairs or quality guarantees.

Keep one habit: Complete trip export → audit counts → height contrast → Data Triage → independent six-row export.

MAIN LESSON · SECTION 1

Inspect the table before trusting a model score

> Count issues without inventing observations or deleting evidence.

Learning goals

I can inspect row meaning **missing value**s and repeated records.

I can document dataset source **schema** and limitations.

I can separate observed quality flags from justified corrections.

Retrieve schema, target and unit reasoning

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.
- Why is elapsed trip duration in minutes a regression target even when recorded as whole numbers?
- How many minutes are 180 seconds?

> **My readiness explanation**

End goal

By the end, I can explain inspect before modeling using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Dataset **shape**, types, **range**s, **duplicate**s, and **missingness** determine whether modeling results can be trusted.



LEARNING TARGET A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

Builder name and experiment date:

Readiness for the connected explanation

Readiness for Units conversions and data provenance

This week's end goal

MAIN LESSON · SECTION 2

Read the complete fictional export and its provenance

> One row is intended to mean one indoor practice trip.

Why inspection comes first

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before **fitting** a model, inspect what one row represents and document where the table came from.

A **dataset card** is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Source and row contract

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes.

Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no **evidence** about outdoor performance.

All five original rows

Trip ID	Duration in minutes	What is visible
A	8	A recorded duration
B	None	An unknown duration
B	None	The same ID appears again
C	-2	A negative duration

Trip ID	Duration in minutes	What is visible
D	12	A recorded duration

Keep flags separate from corrections

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition.

We need **evidence** before deciding what happened.

▼ Optional extension · Convert known source units

Two fictional activity logs use different duration units. Log A records 180 seconds and Log B records 4 minutes.

Keep the **raw value**s, units and source labels; derive a shared unit so another learner can reproduce the comparison.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use inspect before modeling to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: Units conversions and data provenance

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MAIN LESSON · SECTION 3

Trace four audit counts without subtracting overlapping flags

> The small script inspects supplied pairs; it is not a complete validator.

Complete audit program

```
rows = [("A", 8), ("B", None), ("B", None),
        ("C", -2), ("D", 12)]
ids = set()
missing = 0
negative = 0
for trip_id, minutes in rows:
    ids.add(trip_id)
    if minutes is None:
        missing = missing + 1
    elif minutes < 0:
        negative = negative + 1
print(len(rows), len(ids), missing, negative)
```

Explain every output and repeated-ID denominator

A set stores each distinct ID once. The **output** is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs.

This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

Turn an observation into a documented action

FINDING	WHAT IT ESTABLISHES	NEXT ACTION
B appears twice	Identity needs review	Check export history
Two None values	2 of 5 rows lack duration	Keep unknown explicit
C has minus 2	Invalid elapsed duration	Check source and units

Flags can overlap: both rows with ID B also have missing duration. Do not add flag counts as if they were separate groups.

State assumptions and omitted checks

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full **schema** validation. It does not check text values, units, source truth or whether IDs refer to real unique events.

A successful run does not certify data quality.

▼ Optional extension · Preserve raw values while converting

```
rows = [{"source": "A", "value": 180, "unit": "s"},
        {"source": "B", "value": 4, "unit": "min"}]
minutes = []
for row in rows:
    if row["unit"] == "s":
        minutes.append(row["value"] / 60)
    elif row["unit"] == "min":
        minutes.append(row["value"])
    else:
        minutes.append(None)
print(minutes)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
A raw	180 seconds	3.0 minutes
B raw	4 minutes	4 minutes

Read each row from left to right and connect the evidence to the stated rule.

The script prints [3.0, 4]. The slash operator returns a floating-point result. The original dictionaries are unchanged. Assume nonnegative numeric **raw value**s have already been validated.

An unsupported unit produces None here and must be reported for review rather than included in arithmetic.

A minute contains 60 seconds. To express 180 seconds in minutes, divide 180 by 60 to get 3 minutes. The other value is already 4 minutes, so leave it at 4. Now both quantities use the same unit and can be compared.

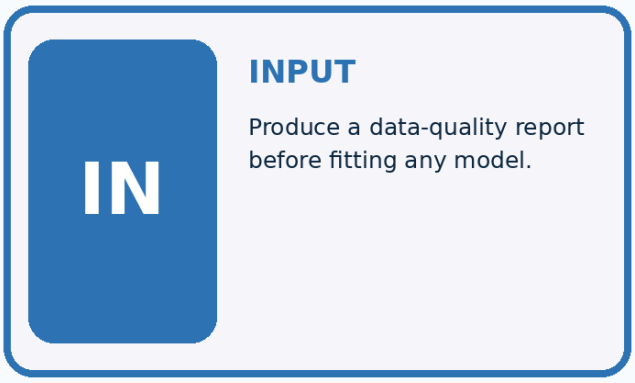
Keep the original number, unit and source beside the converted number. That record lets another person check whether you converted the correct measurement in the correct direction.

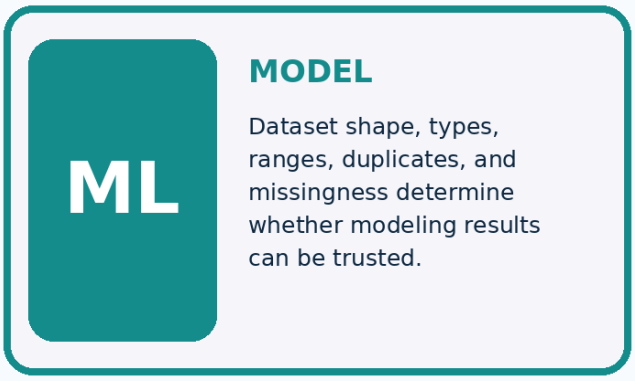
▼ **Earlier wording, alternatives and picture transcripts**

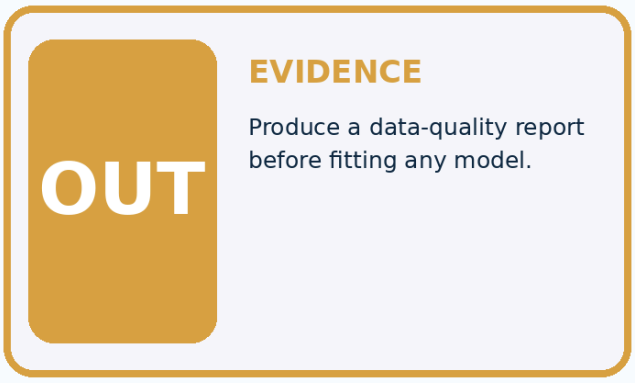
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Produce a data-quality report before **fitting** any model.

MODEL IDEA Dataset **shape**, types, **range**s, **duplicate**s, and **missingness** determine whether modeling results can be trusted.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Dataset **shape** is 40 rows × 5 columns. Inspection finds 3 **duplicate** rows and 4 missing light values.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Apply the connected idea: Units conversions and data provenance

Convert before comparing

Connect the picture and the explanation

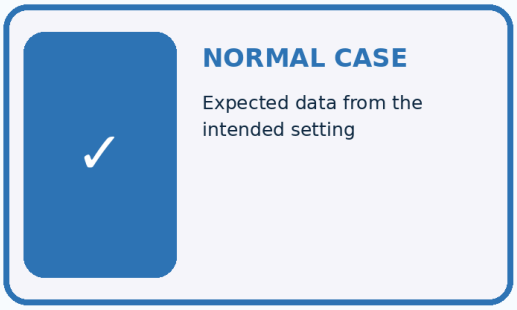
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

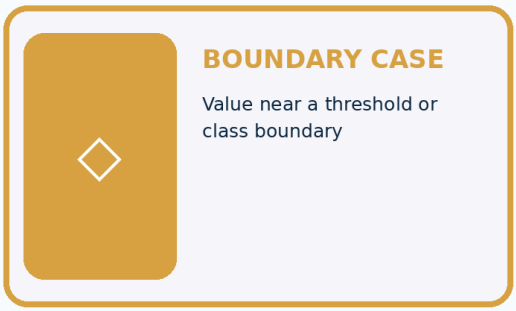
Compare clean-looking values with documented evidence

> Missingness, repeated identity and unit problems require different investigations.

Four evaluation conditions



NORMAL Expected data and expected behavior.

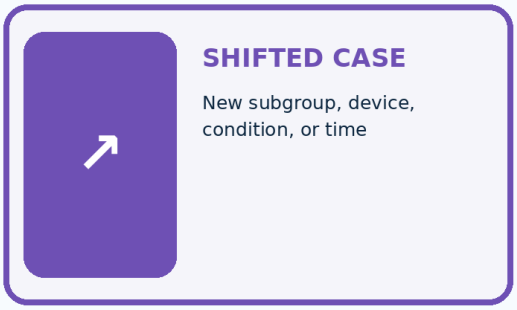


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The teacher confirms B's second row is an accidental **duplicate**.

▼ **Check after predicting**

Prediction: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion.

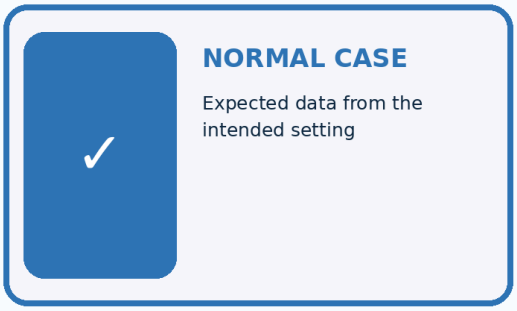
Reason: Resolving one defect does not resolve a separate impossible measurement.

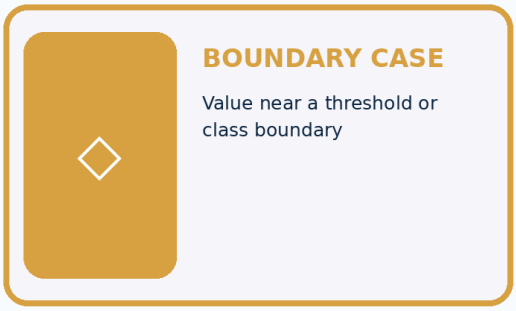
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

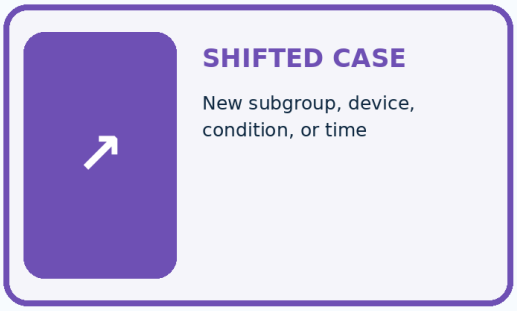
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Follow the height flags, then document the trip export

> A dataset card records findings and limitations rather than certifying quality.

Complete height records and scope

Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.

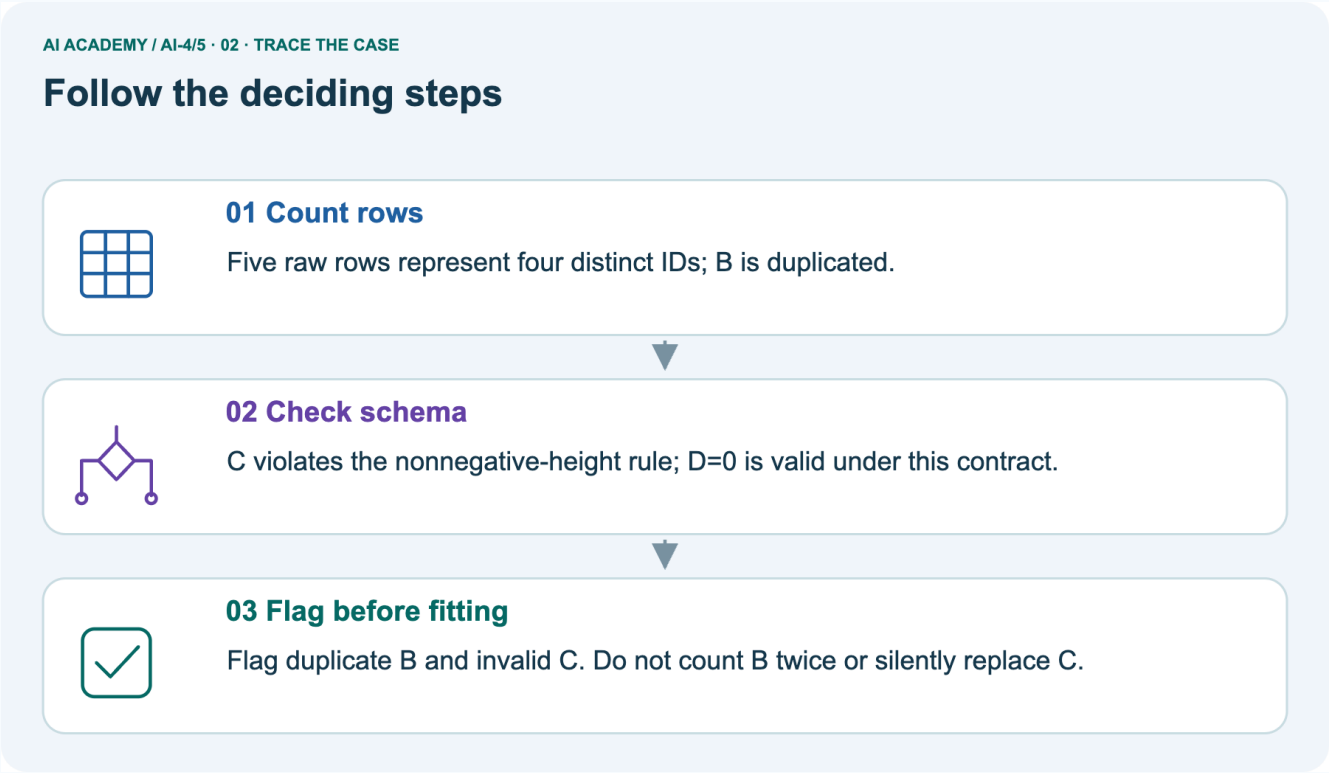
Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep the examples separate. Robot practice v1 has five raw rows, four distinct IDs, two missing-duration rows and one negative row; the extra B occurrence overlaps missingness. The pictured height table instead has A=3,B=5,B=5,C=-2,D=0 cm and no missing values. Its duplicate and invalid-value issues are separate.

A confirmed accidental copy can be removed only in a documented new version; resolving B does not repair C. Zero is allowed by both stated contracts, although validity does not establish measurement truth.

Trace row counts and separate flags



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count rows	Five raw rows represent four distinct IDs; B is duplicated.
2. Check schema	C violates the nonnegative-height rule; D=0 is valid under this contract.
3. Flag before fitting	Flag duplicate B and invalid C. Do not count B twice or silently replace C.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

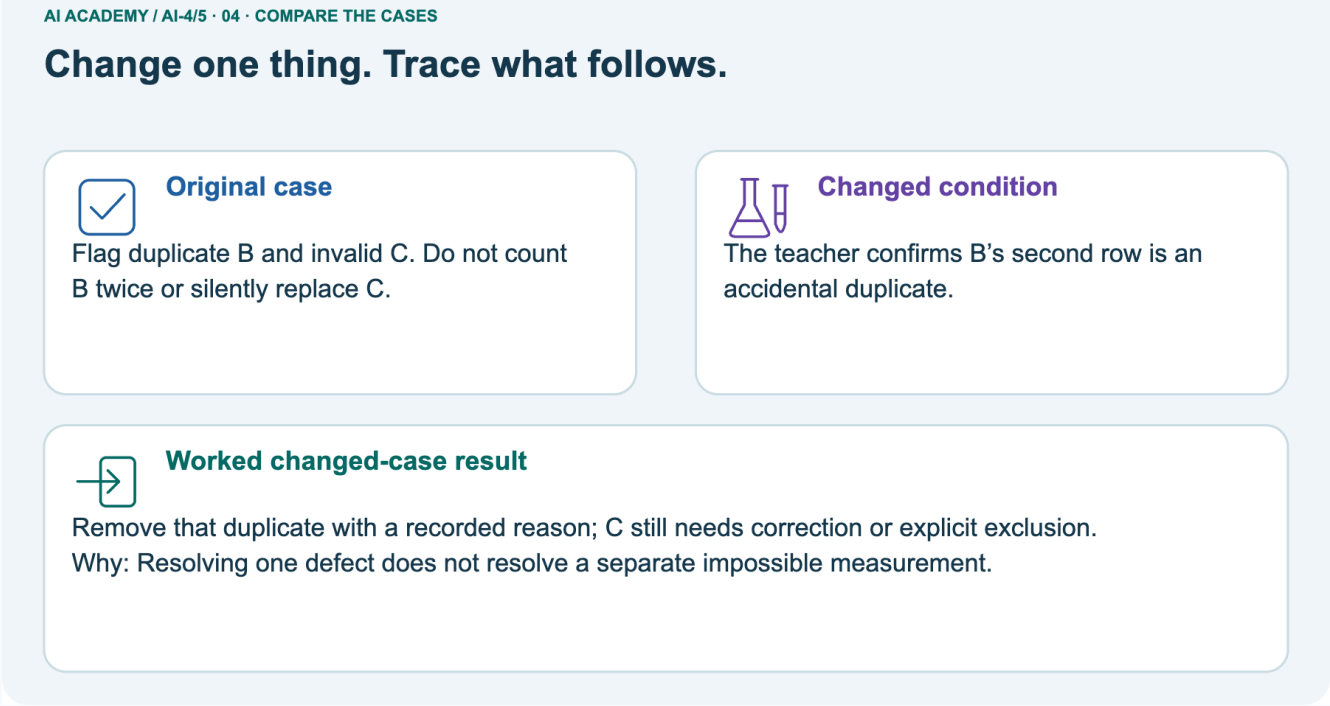
Read the picture in order. Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict a confirmed duplicate correction

- Change one thing: The teacher confirms B's second row is an accidental **duplicate**. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State what remains unknown

- A plausible nonnegative value can still be incorrectly measured; **schema** checks do not establish truth.

- Read the labelled picture: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Write the trip dataset card

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip.

Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the **duplicate**-ID and invalid-duration findings.

State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add intended use, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice.

Do not describe it as validated navigation data.

For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

> **What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.**

▼ **Optional conversion prediction**

> **Convert 90 seconds and 0 seconds. Which is longer: 90 seconds or 2 minutes?**

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 5 • WORKED EVIDENCE

1. Dataset shape is 40 rows × 5 columns.
2. Inspection finds 3 duplicate rows and 4 missing light values.
3. Leaf widths range from 0.4 to 48 cm; inspect the extreme value and unit.
4. Document, repair, or exclude by a stated rule—never silently delete inconvenient rows.

STEP 1 Dataset **shape** is 40 rows × 5 columns.

STEP 2 Inspection finds 3 **duplicate** rows and 4 missing light values.

STEP 3 Leaf widths **range** from 0.4 to 48 cm; inspect the extreme value and unit.

STEP 4 Document, repair, or exclude by a stated rule—never silently delete inconvenient rows.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

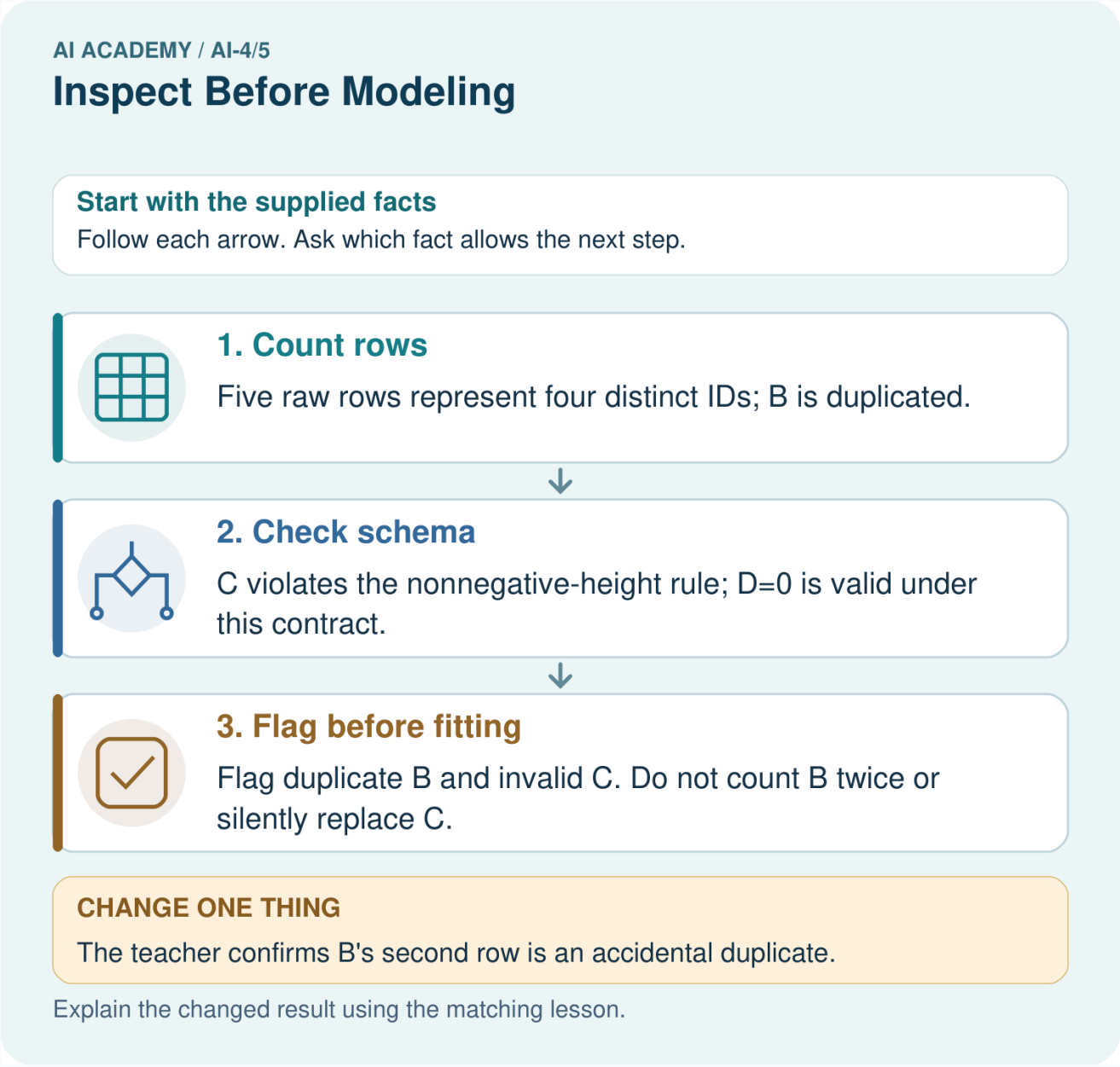
Apply the connected idea: Units conversions and data provenance

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Inspect Before Modeling

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Challenge cleaning without supporting evidence

> Identify the earliest unsupported change or quality claim.

Diagnose the first issue

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

Explain the main limitation

- A complete-looking card does not prove the table is accurate or suitable for a model. It makes claims and unanswered questions reviewable.

  Report raw counts separately from any later accepted subset, with explicit rules for missing, invalid and repeated records. Do not silently change denominators after cleaning.

- For this exercise, zero is allowed and unknown is not zero. That is a stated **schema** choice, not a universal rule for all duration datasets.

  In a real task, measurement resolution and event definitions determine which values make sense. Dataset inspection also does not replace protecting held out data from model choices.

> **A card says the dataset is ready because it has no empty cells. Name two checks that claim misses, then write a more accurate statement about Robot practice v1.**

Challenge the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A model will automatically ignore harmless data-quality problems.

▼ Optional unknown-unit diagnosis

Correct **conversion** does not establish that the sources measured the same activity, used the same start and stop rules, or are accurate. **Provenance** helps someone investigate those differences.

> **A record contains value 8 and unit ticks with no conversion rule. Should you assume seconds?**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A model will automatically ignore harmless data-quality problems. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

A closer explanation

Apply the connected idea: Units conversions and data provenance

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MAIN LESSON · SECTION 8

Explain overlapping counts and avoid invented repairs

> Attempt the audit-count W2 before looking at worked explanations.

Try W2 and explain each count

Use workbook W2 for the connected example “Dataset inspection and cards”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Inspect proposed replacements and coverage claims

A learner replaces both None values with 0 and removes the negative sign from -2. The table now passes a simple nonnegative check, but those edits invent durations.

Keep the original values, record the reason for each flag, and check the source. If a correction is supported, save a new version with the original value, corrected value and reason. If not, keep the uncertainty visible.

Another learner says all robot trips are covered because every row says indoor. A **dataset card** should instead state that outdoor trips are absent and that the data is fictional.

Extra copies of existing rows do not create new conditions or new independent **evidence**.

Check the complete-card claim

- Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain which raw information stays visible

- Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the fresh export

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Optional unit-conversion practice

Comparing raw 180 to raw 4 ignores their units. Dividing 4 by 60 without checking its unit is also wrong: it was already minutes. Branch on the documented unit and keep a **conversion** note.

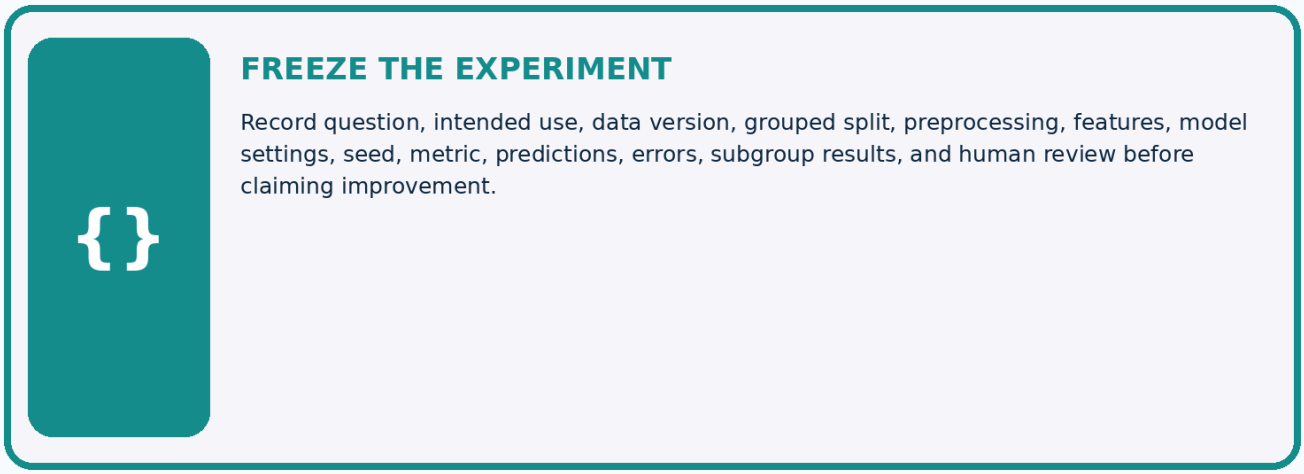
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Apply the connected idea: Units conversions and data provenance

Practise with less help: Interpret the audit counts

Check your reasoning

Practise the connected topic: Units conversions and data provenance

Explain the picture

MAIN LESSON · SECTION 9

Run Data Triage on supplied records and complete probes

> Report issue counts and their overlap before proposing an action.

Complete normal, boundary, missing, unit and invalid probes

Complete Data Triage probe	Required record
Normal (A,8 minutes), intended one row per trip	No visible missing/range issue; source truth still unverified.
Boundary (D,0 minutes), same nonnegative-duration contract	Accept the value under the declared range; do not label zero missing.
Missing (B,None) repeated twice, raw trip export	Flag both rows as unknown and their repeated ID; investigate the source rather than replace with zero.
Changed unit: (U,120 seconds), documented numeric nonnegative elapsed duration	Preserve raw120/unit s; derive2 minutes in a new documented field/version. An unknown unit cannot be guessed.

Complete Data Triage probe	Required record
Invalid (C,-2 minutes), original source unconfirmed	Flag the negative elapsed duration; do not remove the sign. Request evidence for correction.

Inspect, mark overlaps and draft the card

1. Use the five Robot practice v1 rows printed earlier as the complete main investigation. Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Mark row-level flags so their overlap is visible.

   For each flag write keep for review, evidence-supported repair in a new version, or explicit exclusion under a justified rule; none of these supplied flags alone proves which correction is true. Draft the dataset card with source, row meaning, unit, raw version, findings, owner and excluded use. The separate unit probe supplies an actual conversion rule.

   Keep the six-row independent export for an unaided first attempt.

Retain the evidence ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>▼ Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.
</details>

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Data Triage: calculate missing counts, flag ranges, find duplicates, and write a keep/repair/exclude reason for each issue.

1. Write the hypothesis and expected **evidence** before running.

2. Data Triage: calculate missing counts, flag **range**s, find **duplicate**s, and write a keep/repair/exclude reason for each issue.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: inspect a fresh six-row export

> Use the specified denominator and show overlaps explicitly.

Complete new records and contract

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

This selected task uses its own six-row fictional indoor export, not the five-row main example or height picture. The schema intends one row per trip; duration is numeric nonnegative minutes or None for unknown. An extra ID occurrence means raw-row count minus distinct-ID count. List overlapping row flags before any unique-record conclusion; do not silently clean the supplied raw version.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after attempting

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Inspect and document independently

Week 5 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Save an inspection report before fitting the project model

Make the raw-to-reviewed history reproducible.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

Attach the unmodified project table, row contract, source/version and a row-level inspection log. Keep issue counts separate from counts of unique affected records. State any evidenced transformation, unresolved source question and excluded use. If no model was trained, record no model evaluation; a complete dataset card alone cannot establish suitability or quality.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Record limits and the human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with findings, provenance and one unresolved question

> Explain what inspection establishes and what still needs evidence.

Retrieve the reasoning

- Close your notes. Explain this goal in your own words: Inspect row meaning **missing value**s and repeated records

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
shape	number of rows and columns	
range	smallest to largest plausible value	
duplicate	repeated example or record	
missingness	where values are absent	

Closer-explanation terms

Word	Meaning
Provenance	Where data came from and how it was created
Schema	The stated meaning types units and rules for fields
Missing value	An unknown or unavailable value rather than a measured zero

Word	Meaning
Dataset card	A concise record of dataset contents creation intended use and limitations

▼ Extension vocabulary

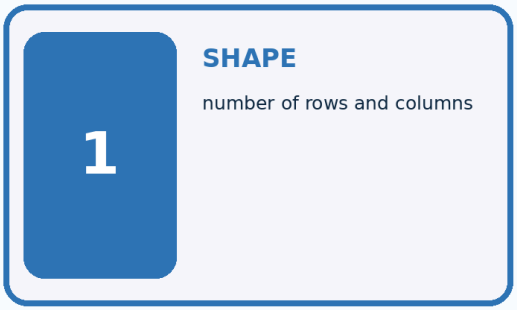
Term	Meaning
Provenance	The origin and processing history of data
Conversion	Expressing a quantity in another unit
Raw value	The value as originally recorded
Derived value	A value calculated from recorded data

▼ Earlier wording, alternatives and picture transcripts

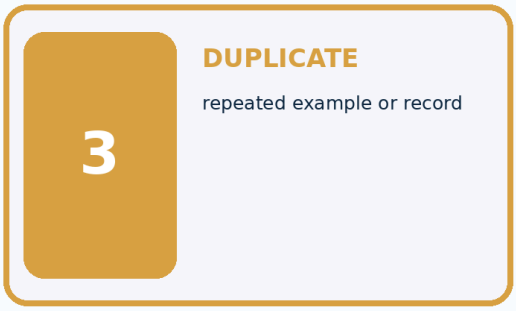
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A model will automatically ignore harmless data-quality problems.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for Units conversions and data provenance

▼ Extra challenge · Build a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **shape**, **range**, **duplicate**, **missingness**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A model will automatically ignore harmless data-quality problems.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

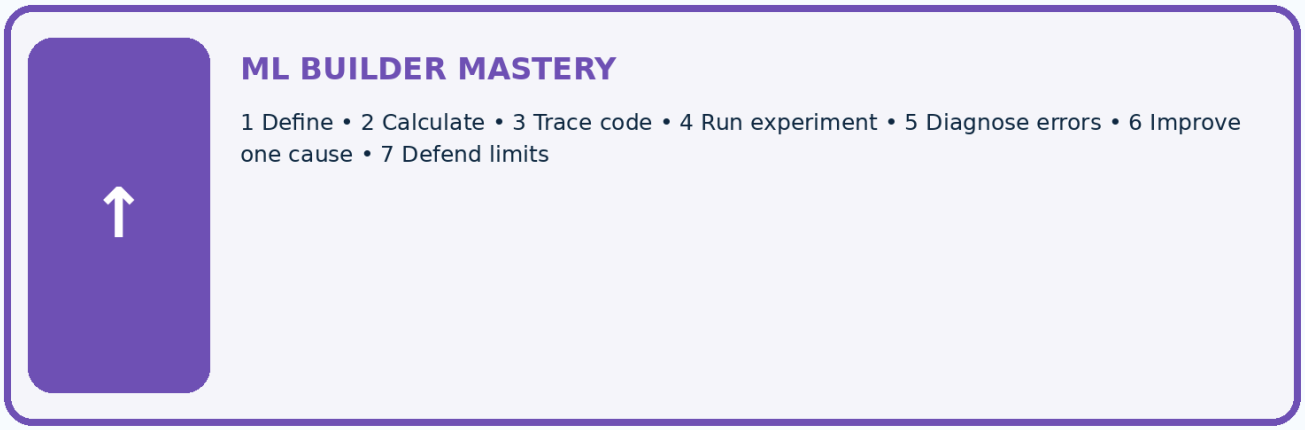
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the fresh six-row indoor export for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.
- Independent **evidence** and defense:

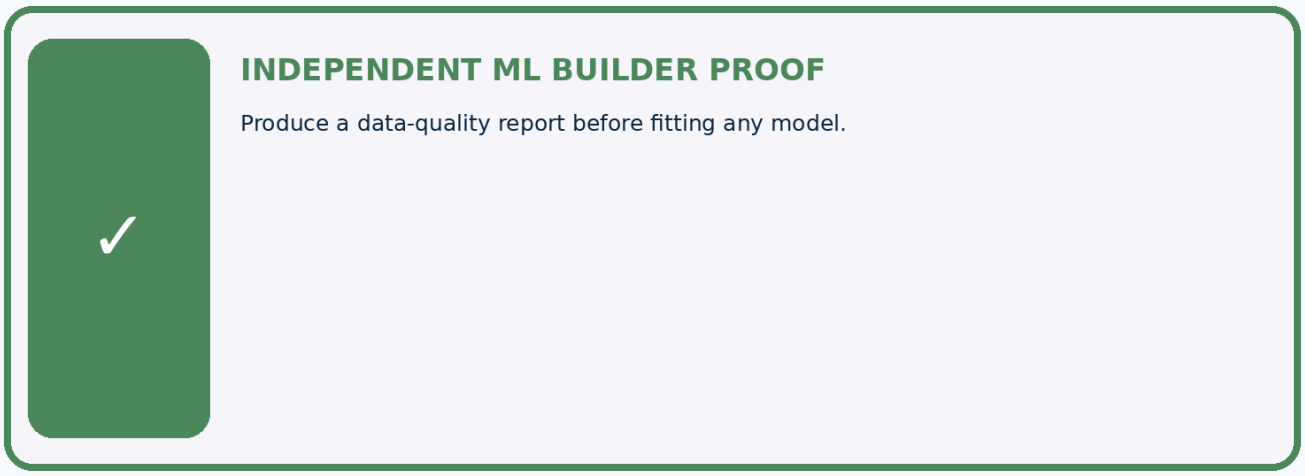
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Produce a data-quality report before **fitting** any model.

NEXT BRIDGE Carry the experiment **evidence** into: Label Quality.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 13

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Lengths include 5 cm, blank and -3 cm, plus a **duplicate** photo row.

Name checks and a deletion risk.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Check **missingness**, physical validity, units and **duplicate** identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Evidence or stage	Value or result
5 cm	Physically plausible
Blank	Missing measurement
−3 cm	Invalid length
Duplicate photo	Check record identity

▼ Earlier wording, alternatives and picture transcripts

Week 5 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Inspect data before choosing repairs

Lengths include 5 cm, blank and -3 cm, plus a **duplicate** photo row.

Name checks and a deletion risk.

Check **missingness**, physical validity, units and **duplicate** identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

▼ **Reference · Earlier route checklist**

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Inspect row meaning missing values and repeated records	Explain the deciding step, show a trace or test, and state one limit.
Document dataset source schema and limitations	Explain the deciding step, show a trace or test, and state one limit.
Separate observed quality flags from justified corrections	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.
- Which labelled step matches this detail? Flag **duplicate** B and invalid C. Do not count B twice or silently replace C.
- 1. Flag before **fitting**
- 2. Count rows
- 3. Check **schema**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Flag before **fitting** Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. Read the picture in order. Flag **duplicate** B and invalid C.

Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The teacher confirms B's second row is an accidental **duplicate**. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

Repair and keep the record

Review Python and Table Readiness (ai-4/4). Goal: Specify a table **schema**. Short practice: Why should the **missing value** not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

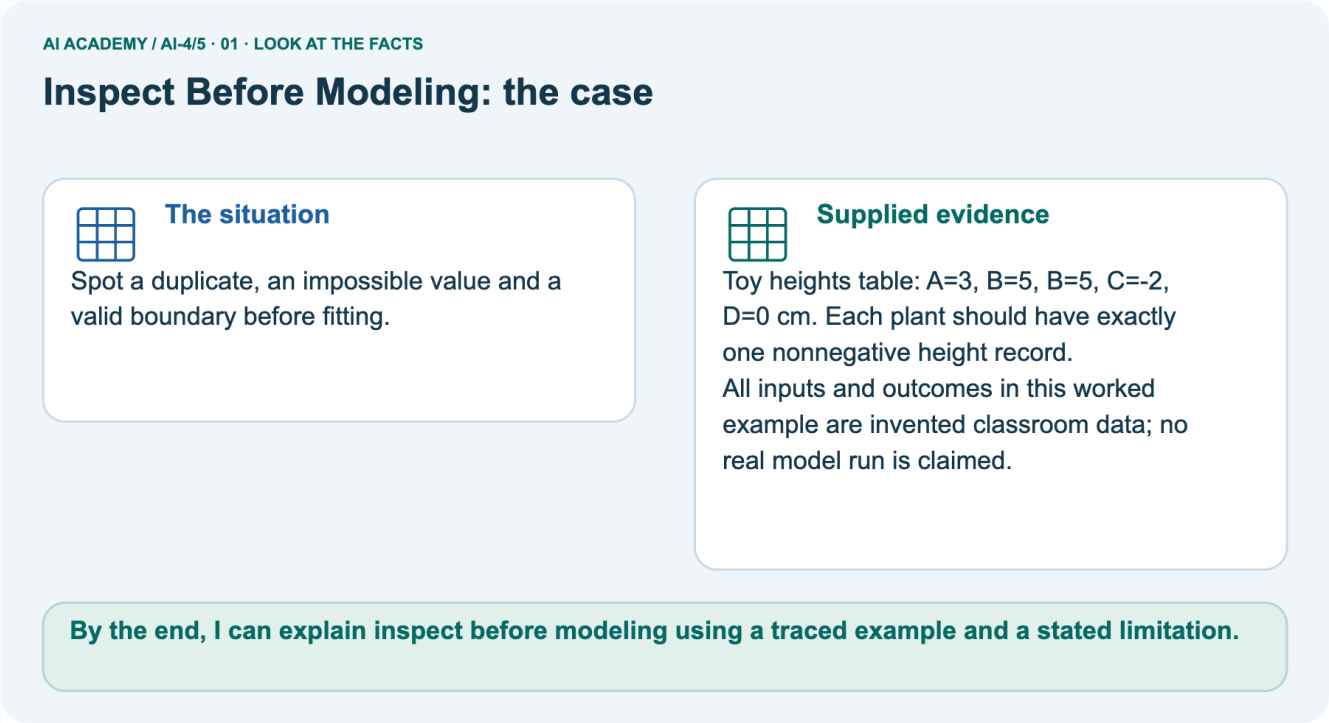
▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Inspect Before Modeling: the case



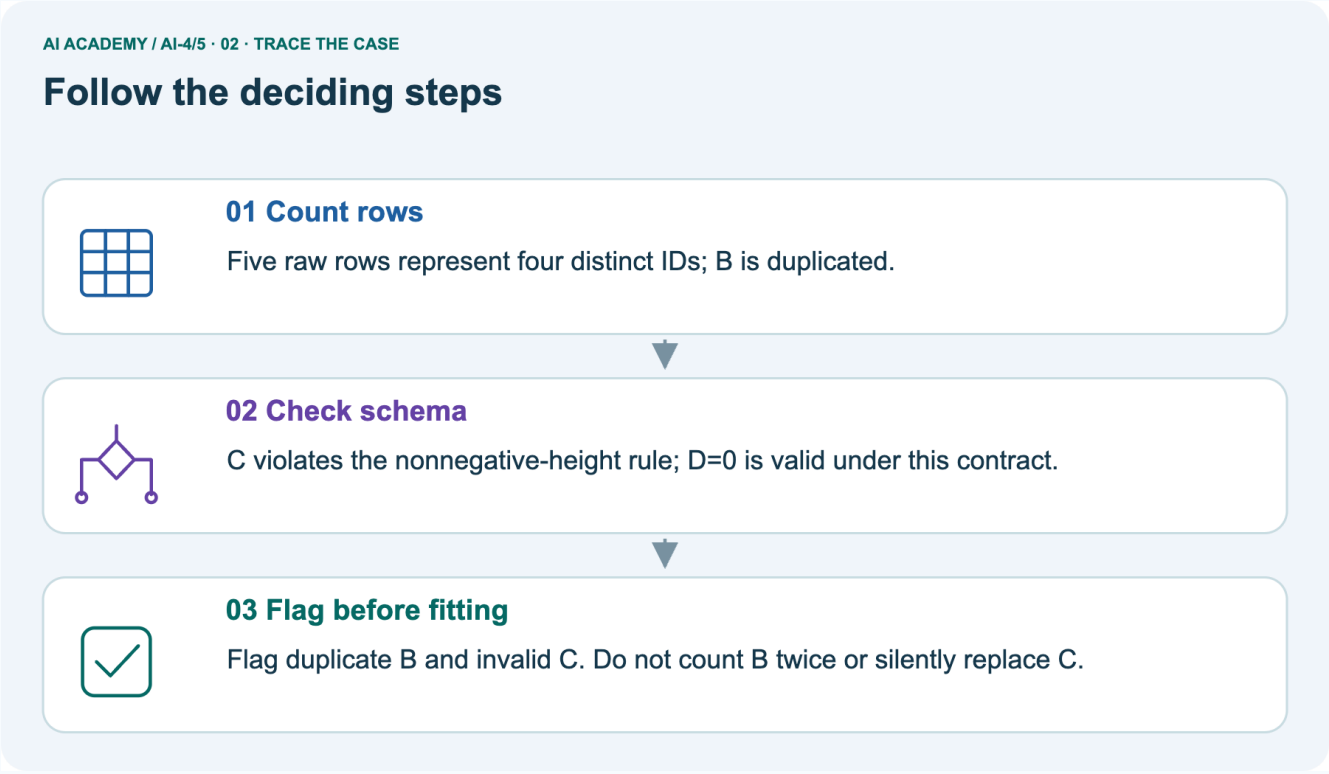
Follow the labelled picture with the matching explanation. Open the vector diagram

Spot a **duplicate**, an impossible value and a valid boundary before **fitting**. Supplied facts: Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



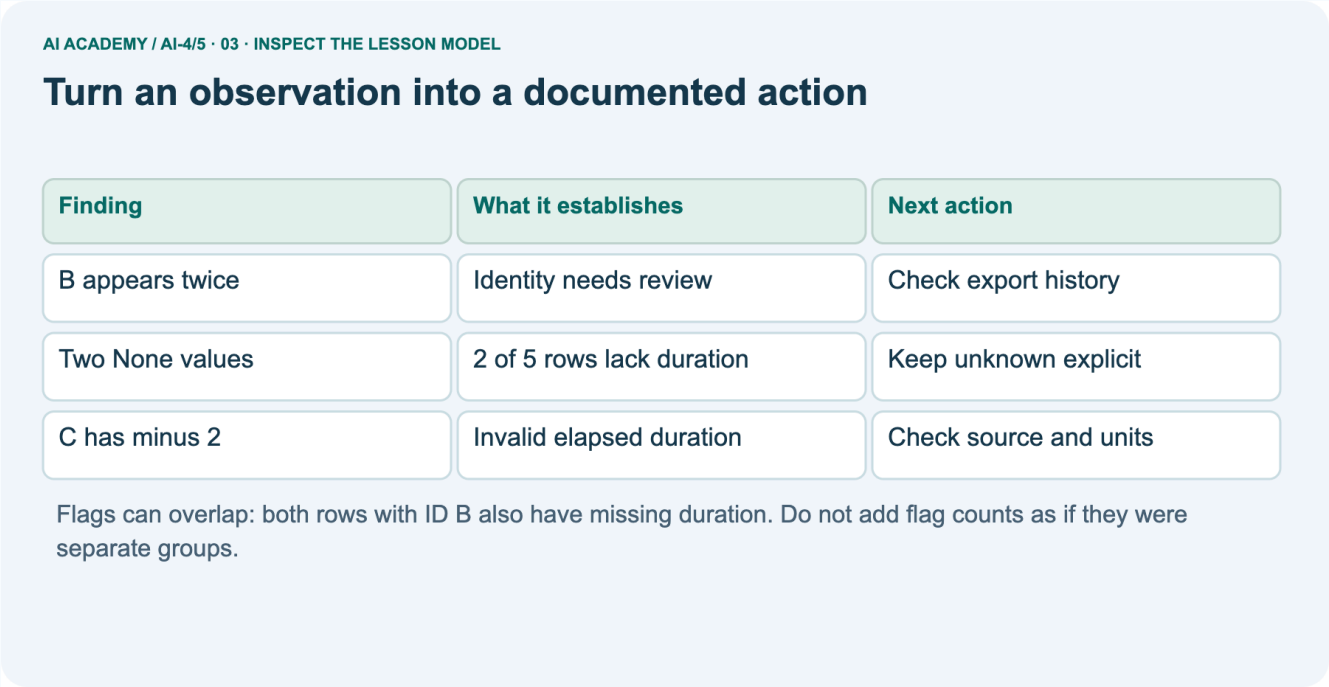
Follow the labelled picture with the matching explanation. Open the vector diagram

Count rows: Five raw rows represent four distinct IDs; B is **duplicate**d. → Check **schema**: C violates the nonnegative-height rule; D=0 is valid under this contract. → Flag before **fitting**: Flag **duplicate** B and invalid C.

Do not count B twice or silently replace C.

Point to each numbered stage and explain which recorded **input** it uses.

03. Turn an observation into a documented action



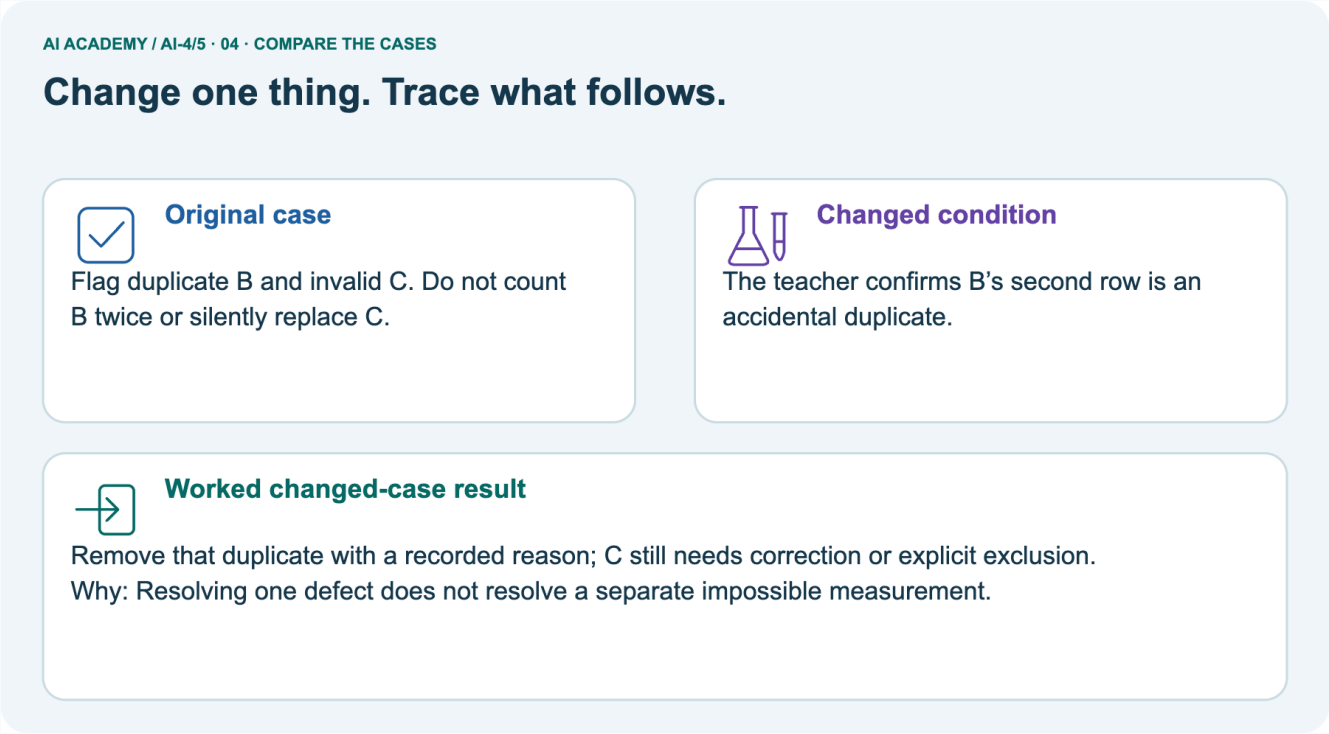
Follow the labelled picture with the matching explanation. Open the vector diagram

Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



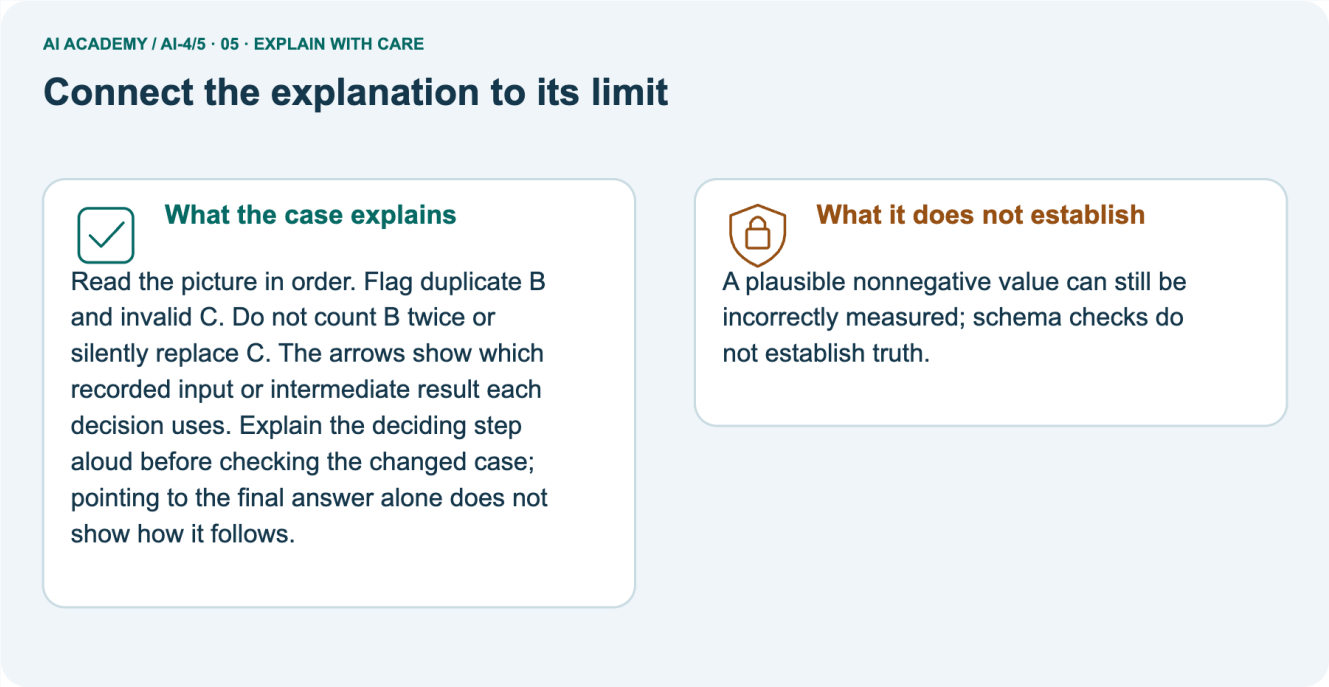
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Flag **duplicate** B and invalid C. Do not count B twice or silently replace C.
Changed condition: The teacher confirms B's second row is an accidental **duplicate**.

Expected result: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Reason: Resolving one defect does not resolve a separate impossible measurement.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

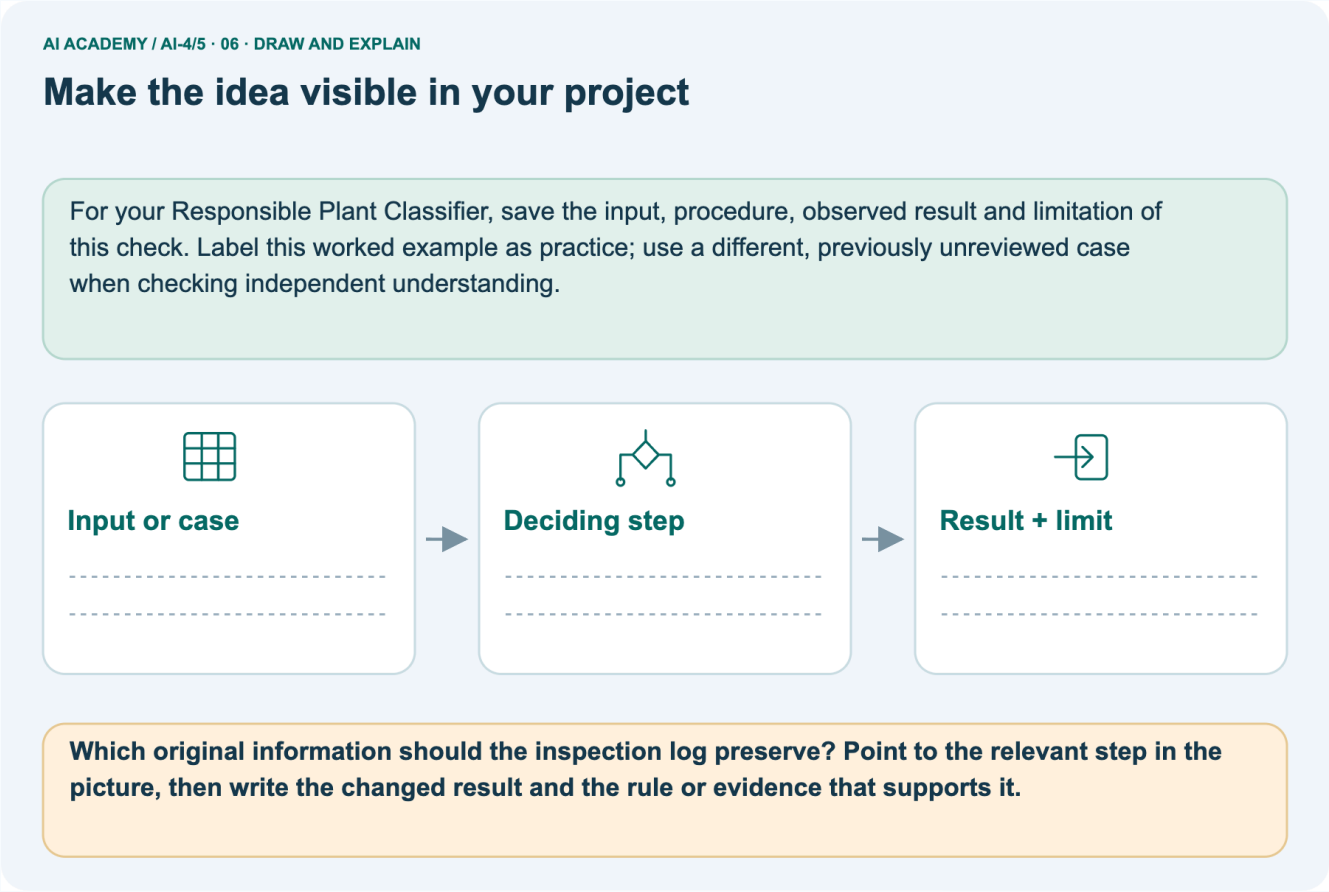
Read the picture in order. Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A plausible nonnegative value can still be incorrectly measured; **schema** checks do not establish truth.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Inspect Before Modeling: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.